llmXive Automated Review of Translation as a Bridging Action: Transferring Manipulation Skills from Humans to Robots

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several strong claims regarding the efficacy of the "bridging action" representation and the scalability of human-only pre-training. While the experimental results generally support the direction of the claims, there are specific instances where the textual descriptions misattribute evidence or overstate the magnitude of the results without sufficient statistical backing.

First, in Section 5.3 (lines 330-335 of `sections/exp.tex`), the authors state: "We also provide the qualitative comparisons in Tab. 5.3, which also shows the superiority of the translation-only bridging action." This is factually incorrect regarding the content of the citation. Table 5.3 (`exp:tab:sec5.3`) exclusively contains quantitative metrics (Progress and Success rates). The qualitative comparisons (visual trajectories) are presented in Figure 5.3 (`exp:fig:sec5.3_traj_frame`) and Figure A.1 (`exp:fig:appdxA_6DoF`). This conflation of evidence types undermines the precision of the claim that the table "shows" qualitative superiority.

Second, in Section 5.6 (lines 430-435 of `sections/exp.tex`), the authors claim that the upper-bound variant "substantially outperforms" the default co-training. While Table 5.6 (`exp:tab:sec5.6`) does show an increase in overall success rate from 38.33% to 55.83%, the term "substantially" is subjective. More critically, the claim implies a uniform improvement, yet the data shows mixed results across task categories (e.g., Microwave tasks show a modest gain, while Drawer tasks show a significant jump). Without a statistical significance test (e.g., t-test or confidence intervals) to confirm that these differences are not due to variance in the 8 trials per task, the strength of the "substantial" claim is not fully supported by the evidence provided.

Finally, in Section 5.4 (lines 360-365 of `sections/exp.tex`), the text asserts that human-only pre-training "substantially improves" data efficiency. While the overall success rate improves from 35.83% to 55.00%, the breakdown in Table 5.4 (`exp:tab:sec5.4`) reveals a counter-intuitive result for "Drawer Tasks," where the success rate actually *decreases* from 43.75% (Stage III only) to 25.00% (Stage I – III). The text glosses over this specific failure case while making a broad claim of improvement. The claim that the method improves efficiency is supported by the aggregate, but the absolute statement ignores the specific task where the pre-training appears detrimental, suggesting the claim should be qualified (e.g., "improves efficiency on average, though with variance across tasks").

These issues are primarily matters of precision in linking claims to their specific evidence and avoiding over-generalization in the presence of mixed results.

Action items.

- [writing] In Sec. 5.3 (exp.tex), the text claims 'We also provide the qualitative comparisons in Tab. 5.3', but Table 5.3 contains only quantitative metrics (Prog./Succ.). The qualitative comparison is in Fig. 5.3 and Fig. A.1. This misattribution confuses the evidence supporting the claim.

- [writing] In Sec. 5.6 (exp.tex), the text states the upper-bound variant 'substantially outperforms' the default co-training. However, Table 5.6 shows the overall success rate increases from 38.33% to 55.83%. While an improvement, the term 'substantially' may be overstated without statistical significance testing or effect size analysis to support the magnitude of the claim.
- [writing] In Sec. 5.4 (exp.tex), the text claims the model 'substantially improves' data efficiency. Table 5.4 shows success rates jumping from 35.83% to 55.00% overall. While a large gain, the claim of 'substantial' improvement for specific task groups (e.g., Drawer tasks where success drops from 43.75% to 25.00%) is not uniformly supported by the data presented in the table.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear and effective teaser diagram that visually summarizes the paper's core concept: transferring human manipulation skills to robots via 'wrist translation' as a bridging action. The layout logically flows from input data to the bridging mechanism and finally to the resulting robot capabilities, with all components clearly labeled and consistent with the caption.

Figure 2

The figure displays the evaluation setups for various tasks, but it fails to include the bottom row of objects mentioned in the caption, creating a discrepancy between the visual content and the description.

Figure 3

Figure 3 effectively communicates the 15 real-world evaluation tasks by pairing clear textual scoring criteria with representative visual examples for each task. The layout is organized by category, and the inclusion of two distinct testing layouts (Scene A and Scene

B) for each task aligns perfectly with the caption's description.

Figure 4

The figure presents the main results but suffers from a confusing legend-to-plot mapping where the two defined metrics (Progress and Success) are interleaved rather than grouped, hindering direct comparison of the methods described in the caption.

Figure 5

The figure effectively displays the per-task performance comparison with clear data labels, but the legend formatting is inconsistent and the x-axis labels are rotated too steeply for optimal readability.

Figure 6

Figure 6 effectively provides a qualitative comparison between the '6DoF Human Actions' and 'Bridging Actions' methods. The side-by-side images clearly illustrate the caption's claim that the bridging action results in a natural pose aligned with the microwave handle, whereas the 6DoF baseline yields a distorted, off-target pose. The visual evidence is clear and directly supports the figure's stated purpose.

Figure 7

The figure fails to communicate the claimed comparison because it lacks any labels or legends to distinguish the 'bridging actions' from the '6DoF baseline' rows, and the visual content does not show the distinct behaviors described in the caption.

Figure 8

Figure 8 effectively visualizes the training loss comparison across three action components (wrist, end-effector, gripper) with and without pre-training. The axes, legends, and error bands are clear, and the plot directly supports the caption's claim that pre-training accelerates convergence for all components.

Figure 9

The figure fails to visualize the claimed data; it displays static task scenes without the projected action vectors or trajectories described in the caption, making the claim of 'close alignment' impossible to verify visually.

Figure 10

The figure presents a per-task comparison but fails to specify the training configuration for the 'Ours' method in the legend, obscuring the specific claim about human-only pre-training. Additionally, the x-axis labels are poorly formatted, leading to overlapping text that hinders readability.

Figure 11

Figure 11 effectively visualizes representative failure trajectories for the 'insert straw' and 'open drawer' tasks. The image clearly labels the specific failure modes ('Fail to grasp the straw', 'Fail to establish a valid pulling contact') in red text, which aligns perfectly with the caption's description of critical moments where the robot fails despite having clear task intent.

Figure 12

The figure provides a clear visual overview of the architecture and data sources, but the caption contains a formatting error ('$_0$-like black2024pi_0') that obscures the model reference.
Action items.

- [science] Figure 2: The caption states 'We show one of the initialized setups for each task (top), and the objects used throughout policy evaluations (bottom),' but the rendered image contains no bottom row; it only displays the top row of task setups.
- [science] Figure 4: The legend defines two metrics ('Prog. (outer)' and 'Succ. (inner)') for each method, but the bars are not grouped or paired by method. Instead, they are interleaved across the x-axis, making it impossible to visually compare the 'Progress' vs 'Success' performance for a specific training stage (e.g., comparing the green bars to the orange bars).
- [writing] Figure 4: The legend uses the term 'w/o human' for the green bars, but the caption describes this as 'Training only on robot pick-and-place data'. While likely equivalent, the legend label is ambiguous and should match the caption's description for clarity.
- [writing] Figure 5: The legend is split into two rows with inconsistent formatting (e.g., 'w/o human' vs 'Co-train (Stage II)'), making it difficult to quickly map colors to the specific training stages described in the caption.
- [writing] Figure 5: The x-axis labels are rotated at a steep angle, which reduces readability and makes it difficult to distinguish between similar task names like 'Stack Left Mug' and 'Stack Right Mug'.
- [science] Figure 7: The caption claims to show a qualitative comparison between 'bridging actions' and a '6DoF baseline', but the rendered image contains no text labels, legends, or color coding to distinguish which rows correspond to which method, making the comparison impossible to verify.
- [writing] Figure 7: The image consists of a grid of identical or near-identical frames of a robot arm and objects with no visible differences in behavior or outcome, failing to illustrate the 'stable manipulation behaviors' claimed in the caption.

- [science] Figure 9: The caption claims to visualize ‘bridging actions’ and ‘6DoF end-effector actions’ projected onto the head camera, but the image contains no visible action vectors, trajectories, or overlays to demonstrate this alignment.
- [writing] Figure 9: The image is a collage of four static scenes with no visual indicators (arrows, lines, or markers) to distinguish between the two action types mentioned in the caption.
- [science] Figure 10: The legend defines ‘Ours’ and ‘Upper Bound’ but does not specify the training data source for the ‘Ours’ bars (e.g., human-only pre-training vs. co-training), making it impossible to verify the caption’s claim about ‘human-only pre-training’ efficiency.
- [writing] Figure 10: The x-axis labels are rotated at a steep angle, causing significant overlap and illegibility for tasks like ‘Wipe Microwave L->R’ and ‘Wipe Microwave R->L’.
- [writing] Figure 12: The caption contains a LaTeX artifact ‘$_0$-like black2024pi_0’ which is likely a broken citation or formatting error; this should be cleaned up for readability.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized robotics and machine learning terminology that creates a barrier for non-specialist readers. The most frequent offender is the use of “embodiment” (Abstract, Intro, Related Work) to simply mean “robot type” or “physical form.” This term is unnecessary jargon that should be replaced with plain English to improve clarity.

Additionally, several acronyms are used without definition at their first occurrence. “6DoF” appears in the Abstract and Introduction without being spelled out as “six degrees of freedom.” Similarly, “VLA” (Vision-Language-Action) is used in the Abstract and Method sections without expansion. While these are standard in the field, the paper’s goal of scaling robot learning suggests a broader potential audience that may not be familiar with these specific abbreviations.

The term “flow matching” is introduced in the Abstract and Method as a standard technique without any descriptive context. A brief explanation of what this technique does (e.g., “a method for generating smooth action sequences”) would help readers understand the methodology without needing to consult external literature.

Finally, the phrase “interleaved action tokens” in the Abstract and Method is technically precise but opaque. Replacing “interleaved” with “mixed” or “combined in a single sequence” would make the architectural contribution more accessible. The term “co-training” is also used frequently; while common, “joint training” is a more intuitive alternative for a general

audience. Addressing these terms will significantly lower the entry barrier for readers outside the immediate robotics sub-field.
Action items.

- [writing] Define '6DoF' at first use in the Abstract and Introduction. While common in robotics, the paper targets a broader audience scaling robot learning, and the acronym is used immediately without expansion (e.g., 'six degrees of freedom').
- [writing] Replace the term 'embodiment' with 'robot type' or 'physical platform' in the Abstract and Introduction. The phrase 'treats humans as just another bi-manual 6DoF embodiment' is jargon-heavy; 'embodiment' is a specific technical term that obscures meaning for non-specialists.
- [writing] Define 'VLA' (Vision-Language-Action) at its first occurrence in the Abstract or Introduction. The paper uses 'VLA model' and 'VLA' repeatedly without spelling out the acronym, assuming reader familiarity with this specific sub-field terminology.
- [writing] Clarify 'flow matching' in the Abstract and Method section. The term is used as a standard technique name, but for a general audience, a brief parenthetical explanation (e.g., 'a generative modeling technique') would improve accessibility.
- [writing] Replace 'co-training' with 'joint training' or 'training together' in the Abstract and Experiments. 'Co-training' is a specific machine learning term that may be confused with semi-supervised learning; 'joint training' is more descriptive for the general reader.
- [writing] Define 'interleaved action tokens' in the Abstract. The phrase is used to describe the model architecture but lacks a plain-language explanation of what 'interleaved' implies in this context (e.g., 'mixed in a single sequence').

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logical argument for using wrist translation as a bridging action to transfer skills from humans to robots, citing noise in 6DoF human pose estimation and the mismatch in contact patterns as primary motivations. The experimental design generally follows this logic, comparing translation-only against 6DoF baselines and demonstrating scalability.

However, there are inconsistencies between the textual claims and the reported quantitative results. In Section 5.3 (Table 5.3), the text asserts the "superiority" of the translation-only bridging action. Yet, the table shows that for "Drawer Tasks," the 6DoF baseline achieves a higher progress score (55.00%) compared to the proposed method (49.06%), although the proposed method has a higher success rate (3.13% vs 0.00%). The claim of general superiority is not fully supported by the mixed results in the table, requiring a more nuanced discussion of where the translation-only approach fails or succeeds relative to the baseline.

Furthermore, the causal link in the "Upper Bound" analysis (Section 5.6) is slightly tenuous. The authors attribute the performance gain in the upper-bound experiment (using robot data with translation-only objectives) to the reduction of "visual gap and action noise." While plausible, the experiment conflates the quality of the data source (robot vs. human) with the action representation. The improvement could be driven primarily by the inherent lower noise and better proprioceptive alignment of the robot data itself, rather than the efficacy of the translation-only representation in isolation. The conclusion that the representation *itself* is the primary driver of the upper bound performance needs to be more carefully qualified.

Finally, the mechanism for the transfer from Stage I (human-only, translation-only) to Stage III (robot, 6DoF) in Section 5.4 is asserted but not fully explained. The paper claims that pre-training on non-executable translation signals improves the efficiency of learning executable 6DoF actions. While the interleaved attention mechanism is described, the logical step explaining *why* optimizing for translation specifically aids the learning of rotation and full 6DoF poses in the absence of rotation supervision during Stage I requires more explicit justification or ablation evidence.
Action items.

- [writing] The paper presents a logical argument for using wrist translation as a bridging action to transfer skills from humans to robots, citing noise in 6DoF human pose estimation and the mismatch in contact patterns as primary motivations. The experimental design generally follows this logic, comparing translation-only against 6DoF baselines and demonstrating scalability. However, there are inconsistencies between the textual claims and the reported quantitative results. In Section 5.3 (Table 5.3), the

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the efficacy of the "bridging action" representation and the scalability of the approach, but the evidence provided in certain sections does not fully support the strength of these assertions.

First, the Abstract and Section 5.1 claim that the proposed method "scales with the amount of human data." The current evidence for this is a binary comparison: a model trained with large-scale human pre-training versus one without. This demonstrates that pre-training is beneficial, but it does not demonstrate *scaling*. To justify the claim of scaling, the authors should ideally present a scaling law analysis (e.g., a plot of performance vs. the logarithm of the number of human hours) showing a consistent trend. Without this, the claim of "scaling" is an extrapolation beyond the provided data points.

Second, the Abstract states that the bridging action transfers skills "far more effectively" than noisy 6DoF human actions. While Table 1 (Sec 5.3) shows a clear improvement in success rate (22.50% vs 12.50%), the absolute success rates for both methods are relatively low. The phrase "far more effectively" is a qualitative superlative that may overstate the practical utility given that the robot still fails the majority of the time. The language should be tempered to reflect the specific quantitative improvement (e.g., "significantly improves success rate") rather than a broad qualitative claim that implies a near-perfect transfer.

Finally, the Upper Bound analysis in Section 5.6 concludes that the results confirm the bridging representation is an "effective medium" for skill transfer. However, the upper bound experiment replaces human data with high-quality robot data, effectively removing the observation gap, action noise, and embodiment mismatch simultaneously. The performance jump from the "Default" to the "Upper Bound" is likely driven more by the removal of these gaps and the higher quality of the robot data than by the specific choice of a translation-only representation. Attributing the success primarily to the representation in this context is an over-interpretation; the experiment validates the potential of the *pipeline* when the data gap is closed, not necessarily the superiority of the translation-only signal in isolation. The conclusion should more carefully distinguish between the benefits of the representation and the benefits of high-fidelity, aligned data.
Action items.

- [science] The claim that the bridging action 'scales with the amount of human data' (Abstract, Sec 5.1) is supported only by a single comparison between a model with and without pre-training. The paper lacks a scaling law analysis (e.g., performance vs. data size curve) to substantiate the 'scales with' assertion. Clarify this claim or provide the missing scaling analysis.
- [writing] The abstract states the method transfers skills 'far more effectively' than 6DoF actions. While Table 1 shows a significant gap in success rate (22.50% vs 12.50%), the term 'far more effectively' is subjective and potentially over-stated given the absolute success rates remain low. Temper the language to reflect the specific quantitative improvement rather than a qualitative superlative.
- [science] In Sec 5.6, the paper claims the upper bound analysis confirms the bridging representation is an 'effective medium' for transfer. However, the upper bound experiment replaces human data with robot data (removing the embodiment gap entirely). This validates the utility of the *data quality* and *visual alignment* more than the specific *translation-only* representation itself. The conclusion over-attributes the performance gain to the representation rather than the removal of the noise/gap.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This review focuses exclusively on safety and ethical considerations regarding the data collection, human subjects, and physical deployment of the robotic system.

Human Subject Data and Consent The manuscript states in Section 3.3 ("Training Strategies") that the authors utilized "approximately 600 hours of human actions," including "~500 hours of out-sourced free-form household manipulation." While the paper mentions using PICO 4 Ultra Enterprise headsets for data collection, it provides no information regarding the ethical oversight of this data collection. Specifically, the text lacks:

1. IRB/Ethics Approval: A statement confirming whether the data collection protocol was reviewed and approved by an Institutional Review Board (IRB) or equivalent ethics committee.
2. Informed Consent: Details on how human participants were informed about the data usage, storage, and potential risks, and whether explicit consent was obtained.
3. Privacy and Anonymity: Given the use of head-mounted cameras (PICO 4) and "in-the-wild" or "out-sourced" settings, there is a high risk of capturing identifiable faces, private environments, or sensitive activities. The paper does not describe the methods used to anonymize this data (e.g., blurring faces, cropping backgrounds) or how privacy was maintained.

Without these disclosures, the reproducibility and ethical validity of the human data component are unclear.

Physical Safety and Risk Mitigation The experiments involve a bi-manual mobile robot (ByteMini) performing manipulation tasks in real-world environments (Section 4.1), including opening microwaves, handling mugs, and interacting with drawers. These tasks carry inherent risks of physical harm to humans (if present), damage to property, or injury to the robot itself.

- The "Evaluation Setups" section (Section 4.1) describes the tasks and metrics but does not mention safety protocols.
- There is no description of emergency stop mechanisms, collision avoidance strategies during the rollout phase, or supervision requirements (e.g., was a human operator present to intervene?).
- The "Failure Case Analysis" (Section 5.8) acknowledges that the robot fails at grasping straws or opening drawers, which could lead to dropped objects or collisions. The paper should explicitly state how these failure modes were managed to ensure safety during the 8 trials per task.

Recommendations To address these concerns, the authors should add a dedicated "Ethics and Safety" subsection or expand the "Experimental Setup" to include:

- Confirmation of IRB approval or a statement that the research was exempt, along with details on consent procedures for the 500- hours of human data.
- A description of data anonymization techniques applied to the video streams.
- A summary of safety measures implemented during real-robot experiments (e.g., safety zones, emergency stops, human supervision).

These additions are necessary to ensure the research adheres to standard ethical guidelines for human-subject research and safe robotic deployment.
Action items.

- [writing] The manuscript describes collecting ~500 hours of 'out-sourced free-form household manipulation' data (Sec. 3.3) but lacks details on human subject consent, IRB approval, or data privacy protocols. Explicitly state the ethical review status and how participant anonymity was preserved.
- [writing] The study involves real-world robot deployment on 15 manipulation tasks (Sec. 4.1) with potential for physical harm (e.g., breaking objects, pinching). The paper does not mention safety protocols, emergency stop mechanisms, or risk mitigation strategies used during data collection and evaluation.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling hypothesis that translation-only wrist actions serve as a robust bridging representation for transferring manipulation skills from humans to robots. The experimental design, which includes a three-stage training strategy and comparisons against 6DoF baselines, is generally sound. However, the scientific evidence supporting the central claims requires strengthening in terms of statistical rigor and quantitative validation of qualitative observations.

First, the primary results presented in Tables 1, 2, 3, and 4 (Sec 5.2–5.6) report only mean success rates and progress scores. Given that the evaluation involves 15 distinct tasks with 8 trials each (Sec 5.1), the absence of standard deviations, confidence intervals, or statistical significance tests (e.g., t-tests or ANOVA) makes it difficult to assess the robustness of the reported improvements. For instance, the jump from 12.50% to 22.50% overall success in Table 1 (Sec 5.3) is notable, but without variance data, it is unclear if this difference is statistically significant or driven by outliers in specific tasks.

Second, the claim of scalability with "large-scale human-only pre-training" (600 hours, Sec 3.3) lacks sufficient detail on the data composition. The evidence would be stronger if the authors reported the number of unique human actors, the diversity of the environments, and the distribution of tasks within this 600-hour dataset. Without this, there is a risk that the performance gains are due to overfitting to a specific subset of human styles or environments rather than genuine generalization.

Finally, the alignment analysis in Sec 5.6 relies heavily on qualitative visualizations (Fig 5.6) to claim that the bridging action aligns with executable robot actions. While the visual evidence is suggestive, the claim that the bridging action is a "reliable alternative" requires quantitative support. The authors should report numerical metrics, such as the mean Euclidean distance between the projected bridging trajectories and the ground-truth 6DoF trajectories, or a correlation coefficient, to substantiate this claim. The current reliance on visual inspection alone is insufficient for a rigorous scientific conclusion.
Action items.

- [science] Report statistical significance (e.g., p-values, confidence intervals, or standard deviations) for the success rate and progress metrics in Tables 1, 2, 3, and 4. The current presentation of single-point estimates across 15 tasks lacks evidence of robustness against variance.
- [science] Clarify the sample size and variance for the 'large-scale human-only pre-training' (600 hours). Specify the number of unique human actors and the distribution of tasks to assess potential bias or overfitting to specific human styles.
- [science] Provide quantitative metrics (e.g., mean squared error or correlation coefficients) for the 'alignment' claims in Sec 5.6 and Fig 5.6. Qualitative visualizations of trajectory alignment are insufficient to substantiate the claim that the bridging action is a 'reliable alternative' without numerical error bounds.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the experimental section (Sec. 5) is currently insufficient to support the strong claims regarding the efficacy of the bridging action representation. While the paper presents clear trends in Tables 1 through 4 (e.g., `exp:tab:sec5.3`, `exp:tab:sec5.4`, `exp:tab:sec5.5`, `exp:tab:sec5.6`), it relies exclusively on point estimates (mean success rates and progress percentages) without any measure of variance, uncertainty, or statistical significance.

Specifically, in Table 1 (`exp:tab:sec5.3`), the authors claim the bridging action is superior to 6DoF actions (22.50% vs 12.50% overall success). Without standard deviations or confidence intervals derived from multiple independent runs (seeds), it is impossible to determine if this 10% gap is statistically significant or a result of random initialization variance, which is common in VLA training. The text mentions "eight trials per task" (Sec 5.1), but it is unclear if the reported metrics are aggregated over these trials or if the experiments were repeated with different random seeds. If the 120 total trials (15 tasks × 8 trials) constitute the entire dataset, the effective sample size for task-level generalization is small (N=15), necessitating a report of per-task variance or a non-parametric test (e.g., Wilcoxon signed-rank) to validate the "substantial improvements" claimed in Sec 5.2.

Furthermore, the ablation studies in Table 3 (`exp:tab:sec5.5`) and Table 4 (`exp:tab:sec5.6`) present single-point comparisons. In deep learning for robotics, performance can fluctuate significantly based on hyperparameters and random seeds. The claim that removing the bridging objective "greatly degrades" performance (dropping from 38.33% to 12.50%) requires evidence that this drop is consistent across multiple training runs. The absence of error bars in the figures (e.g., `exp:fig:sec5.2_results`, `exp:fig:appdxB_main_results`) and the lack of statistical tests (t-tests, ANOVA, or confidence intervals) in the text prevents a rigorous assessment of the results' reliability.

To meet the standards of statistical rigor expected for this venue, the authors must:

1. Report results as mean ± standard deviation over at least 3-5 independent training seeds.
2. Include p-values or confidence intervals for the key comparisons in Tables 1-4.
3. Clarify whether the "eight trials per task" refers to evaluation rollouts only, and if so, explicitly state the number of training seeds used to generate the reported means.

Without these additions, the quantitative evidence remains suggestive but not statistically conclusive.

Action items.

- [science] Report statistical significance (e.g., p-values, confidence intervals, or standard deviations) for the success rate and progress metrics in Tables 1, 2, 3, and 4. Currently, single point estimates are presented without error bars or variance measures, making it impossible to assess if the observed improvements (e.g., 22.50% vs 12.50% in Tab 1) are statistically robust or due to random variation.
- [science] Clarify the statistical protocol for the 15 tasks. The text states 'eight trials per task' (Sec 5.1), but the tables report aggregate percentages. Specify if these percentages are averages over the 120 total trials (15 tasks * 8 trials) or if they represent the proportion of tasks where success was achieved. If the latter, the sample size (N=15) is too small for robust statistical inference without reporting variance across tasks.
- [science] The ablation study in Table 3 (Sec 5.5) compares two models. Given the high variance often seen in robotic learning, a single run comparison is insufficient. Please confirm if these results are averaged over multiple random seeds or independent training runs, and report the standard deviation to validate the claim that the performance drop is 'essential' rather than a stochastic fluctuation.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript is generally well-structured and readable, with a clear logical flow from the problem statement to the proposed solution and experimental validation. The abstract effectively summarizes the core contribution, and the introduction sets the stage well. However, there are several specific instances of grammatical errors and imprecise phrasing that detract from the professional polish of the paper.

In the Experiments section, there are notable inconsistencies between text descriptions and the referenced content. Specifically, in Section 5.3, the text claims to provide "qualitative comparisons in Tab.~\ref{exp:tab:sec5.3}," but the table actually presents quantitative success rates and progress metrics. The qualitative analysis is located in the figures, not the table. This discrepancy forces the reader to search for the wrong type of evidence. Similarly, in Section 5.4, the phrase "no robot trajectories is involved" contains a subject-verb agreement error ("trajectories" is plural, requiring "are").

Grammar and syntax issues also appear in Section 5.6, where the phrase "base on the same vision" should be corrected to "based on." Additionally, in Section 5.2, the sentence "The bridging action can also benefit from large-scale human-only pre-training" is semantically slightly off; actions themselves do not benefit from pre-training, but rather the policy model trained with those actions does. Refining this to "The model trained with the bridging action benefits..." would improve clarity.

Finally, the Conclusion section repeats the phrase "In this work" and "We propose" in close proximity to the Introduction's summary, which is acceptable but could be tightened for better flow. Addressing these specific grammatical slips and reference mismatches will significantly enhance the overall readability and credibility of the manuscript.

Action items.

- [writing] In Section 5.3 (exp.tex), the sentence 'We also provide the qualitative comparisons in Tab.~\ref{exp:tab:sec5.3}' is factually incorrect regarding the content type. Table 5.3 contains quantitative metrics (Progress/Succ %), while qualitative comparisons are in Figures 5.3 and AppdxA. This mislabeling confuses the reader.
- [writing] In Section 5.6 (exp.tex), the phrase 'produce bridging actions and end-effector actions base on the same vision' contains a grammatical error. 'Base' should be corrected to 'based' to maintain standard English usage.
- [writing] In Section 5.4 (exp.tex), the sentence 'no robot trajectories is involved' contains a subject-verb agreement error. 'Trajectories' is plural, so the verb should be 'are' (i.e., 'no robot trajectories are involved').
- [writing] In Section 5.2 (exp.tex), the phrase 'The bridging action can also benefit from

large-scale human-only pre-training' is slightly ambiguous. It implies the action itself benefits, whereas the model utilizing the action benefits. Consider rephrasing to 'The model utilizing the bridging action benefits...' for precision.